

## Supporting Information

Cecala, J.M. and Wilson Rankin, E.E. Wild bee functional diversity and plant associations in native and conventional plant nurseries.

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**Table S1.** The 13 plant nurseries sampled and their physical characteristics. Also included are the total numbers of individuals and species of bees collected by sweep netting and vane trapping, respectively, at each nursery.

| <b>Nursery</b> | <b>Type</b>  | <b>County<br/>(California,<br/>USA)</b> | <b>Area<br/>(ha)</b> | <b>Number<br/>of plots</b> | <b>Proportion<br/>natural area<br/>(1-km buffer)</b> | <b>Elevation<br/>(m)</b> | <b>Sweep netted<br/>bees</b> | <b>Vane trapped<br/>bees</b> |
|----------------|--------------|-----------------------------------------|----------------------|----------------------------|------------------------------------------------------|--------------------------|------------------------------|------------------------------|
| <b>01</b>      | conventional | Riverside                               | 4.00                 | 4                          | 0.257                                                | 316                      | 61 bees in 19 spp.           | 1,437 bees in 28 spp.        |
| <b>02</b>      | conventional | Riverside                               | 2.90                 | 3                          | 0.210                                                | 303                      | 44 bees in 18 spp.           | 524 bees in 26 spp.          |
| <b>03</b>      | conventional | Riverside                               | 3.89                 | 4                          | 0.289                                                | 580                      | 7 bees in 5 spp.             | 245 bees in 22 spp.          |
| <b>04</b>      | conventional | Riverside                               | 3.80                 | 4                          | 0.081                                                | 523                      | 0 bees                       | 17 bees in 8 spp.            |
| <b>05</b>      | conventional | Riverside                               | 30.00                | 12                         | 0.562                                                | 471                      | 21 bees in 10 spp.           | 1,098 bees in 41 spp.        |
| <b>06</b>      | native       | San Diego                               | 0.50                 | 2                          | 0.711                                                | 105                      | 5 bees in 4 spp.             | 5 bees in 5 spp.             |
| <b>07</b>      | native       | San Diego                               | 4.83                 | 4                          | 0.915                                                | 471                      | 50 bees in 23 spp.           | 386 bees in 37 spp.          |
| <b>08</b>      | native       | Orange                                  | 7.30                 | 7                          | 0.793                                                | 114                      | 42 bees in 23 spp.           | 751 bees in 53 spp.          |
| <b>09</b>      | conventional | Los Angeles                             | 4.58                 | 4                          | 0.106                                                | 151                      | 44 bees in 21 spp.           | 1,004 bees in 37 spp.        |
| <b>10</b>      | conventional | Los Angeles                             | 4.29                 | 4                          | 0.151                                                | 104                      | 10 bees in 8 spp.            | 147 bees in 20 spp.          |
| <b>11</b>      | native       | Los Angeles                             | 3.10                 | 3                          | 0.301                                                | 303                      | 52 bees in 29 spp.           | 185 bees in 29 spp.          |
| <b>12</b>      | native       | San Diego                               | 7.21                 | 7                          | 0.300                                                | 12                       | 22 bees in 12 spp.           | 319 bees in 22 spp.          |
| <b>13</b>      | conventional | Riverside                               | 3.86                 | 4                          | 0.614                                                | 775                      | 17 bees in 12 spp.           | 194 bees in 30 spp.          |
| <b>TOTAL</b>   |              |                                         |                      |                            |                                                      |                          | 375 bees in 92 spp.          | 6,312 bees in 102 spp.       |

**Table S2.** Sampling schedule for the 13 nurseries in the study from March 2016 to May 2018. A “1” indicates that nursery was sampled, while a “0” indicates it was not sampled due to logistical reasons. Blacked-out cells indicate seasons when sampling was not possible due to closure of nursery property. Columns sums (bottom row) indicate the total number of nurseries sampled in that season, while row sums (rightmost column) indicate the total number of times that nursery was sampled in the study.

| Nursery           | 2016   |        |        | 2017   |        |        | 2018   | samplings per nursery |
|-------------------|--------|--------|--------|--------|--------|--------|--------|-----------------------|
|                   | Spring | Summer | Autumn | Spring | Summer | Autumn | Spring |                       |
| 01                | 1      | 1      | 1      | 1      | 1      | 1      | 1      | 7                     |
| 02                | 1      | 1      | 1      | 1      | 1      | 1      | 1      | 7                     |
| 03                | 1      | 1      |        |        |        |        |        | 2                     |
| 04                | 1      | 0      | 0      | 1      |        |        |        | 2                     |
| 05                | 1      | 1      | 1      | 1      | 1      | 0      | 0      | 5                     |
| 06                | 1      |        |        |        |        |        |        | 1                     |
| 07                | 1      | 1      | 1      | 1      | 1      | 0      | 1      | 6                     |
| 08                | 0      | 1      | 1      | 1      | 1      | 1      | 1      | 6                     |
| 09                |        | 1      | 1      | 1      | 1      | 1      | 1      | 6                     |
| 10                | 0      | 1      | 1      | 1      | 1      | 1      | 1      | 6                     |
| 11                | 0      | 1      | 1      | 1      | 1      | 0      | 1      | 5                     |
| 12                | 0      | 0      | 0      | 1      | 1      | 0      | 1      | 3                     |
| 13                | 0      | 0      | 0      | 0      | 1      | 0      | 1      | 2                     |
| nurseries sampled | 7      | 9      | 8      | 10     | 10     | 5      | 9      | 58                    |

**Table S3.** List of bee species and morphotaxa, ordered by family and genus, collected in this study via sweep netting and blue vane traps. Morphotaxa are indicated by a species epithet consisting of “sp.” followed by a unique identifier. Each species’ diet breadth, nesting location, and social organization is designated along with the corresponding literature reference as a bracketed number. Species-level information on functional traits was not available for all species in the study, or, of course, for morphotaxa. In such cases, functional traits were assigned based on inference from their taxonomy at the genus level (e.g., Michener (2007) notes that “all *Andrena* species nest in the ground”). Instances where functional traits were assigned in this manner are indicated by table cells merged across species within the column designating the trait. Lastly, the numbers of individual bees (by sex and collection method) collected in the study are listed at the right. In the “subgenus” column: [unk] = taxon was not identifiable to a described subgenus; [NA] = subgenera are not currently broadly established for this genus.

| Taxonomy   |            |              |                        | Functional traits |                  |                     | sweep netting |       | vane trapping |       |
|------------|------------|--------------|------------------------|-------------------|------------------|---------------------|---------------|-------|---------------|-------|
| family     | genus      | subgenus     | species                | diet breadth      | nesting location | social organization | females       | males | females       | males |
| Andrenidae | Andrena    | Plastandrena | Andrena prunorum       | generalist[1]     | below ground[2]  | solitary[2]         | 0             | 1     | 0             | 5     |
|            |            | Simandrena   | Andrena angustitarsata | generalist[3]     |                  |                     | 1             | 0     | 1             | 0     |
|            |            | [unk]        | Andrena sp. 2c-A       | specialist[4]     |                  |                     | 0             | 0     | 5             | 1     |
|            |            | [unk]        | Andrena sp. 2c-B       |                   |                  |                     | 0             | 0     | 2             | 0     |
|            |            | [unk]        | Andrena sp. 3c-B       |                   |                  |                     | 0             | 0     | 2             | 0     |
|            |            | [unk]        | Andrena sp. 3c-C       |                   |                  |                     | 0             | 0     | 1             | 2     |
|            |            | [unk]        | Andrena sp. 3c-D       |                   |                  |                     | 0             | 10    | 0             | 2     |
|            | Calliopsis | Calliopsis   | Calliopsis rhodophila  | generalist[5]     | below ground[2]  | solitary[6]         | 4             | 1     | 11            | 1     |
|            |            | Nomadopsis   | Calliopsis scutellaris | generalist[3]     |                  |                     | 2             | 1     | 0             | 0     |
|            | Panurginus | [NA]         | Panurginus bilobatus   | specialist[3]     | below ground[2]  | solitary[6]         | 2             | 1     | 5             | 3     |
|            |            |              | Panurginus sp. B       |                   |                  |                     | 0             | 0     | 1             | 0     |
|            | Perdita    | Perdita      | Perdita rhois          | generalist[3]     | below ground[2]  | solitary[6]         | 1             | 5     | 0             | 0     |
|            |            | Pygoperdita  | Perdita interrupta     | specialist[3]     |                  |                     | 1             | 4     | 0             | 0     |
|            |            | Pygoperdita  | Perdita micheneri      | specialist[2]     |                  |                     | 3             | 1     | 1             | 2     |
|            |            | Pygoperdita  | Perdita mojavensis     | specialist[3]     |                  |                     | 0             | 0     | 1             | 1     |

|        |                     |                         |                                 |                |                 |             |    |   |     |    |
|--------|---------------------|-------------------------|---------------------------------|----------------|-----------------|-------------|----|---|-----|----|
|        |                     | [unk]                   | <i>Perdita</i> sp. B            | specialist[2]  |                 |             | 0  | 0 | 2   | 3  |
|        |                     | [unk]                   | <i>Perdita</i> sp. D            |                |                 |             | 0  | 0 | 4   | 0  |
|        |                     | [unk]                   | <i>Perdita</i> sp. E            |                |                 |             | 0  | 0 | 2   | 0  |
|        |                     | [unk]                   | <i>Perdita</i> sp. F            |                |                 |             | 2  | 0 | 5   | 3  |
| Apidae | <i>Anthophora</i>   | <i>Anthophoroides</i>   | <i>Anthophora californica</i>   | generalist [7] | below ground[7] | solitary[2] | 9  | 7 | 4   | 7  |
|        |                     | <i>Micranthophora</i>   | <i>Anthophora curta</i>         | generalist[3]  | below ground[2] | solitary[2] | 1  | 0 | 0   | 0  |
|        |                     | <i>Mystacanthophora</i> | <i>Anthophora urbana</i>        | generalist[3]  |                 |             | 6  | 0 | 126 | 10 |
|        |                     | [unk]                   | <i>Anthophora</i> sp. A         | generalist[3]  |                 |             | 1  | 0 | 0   | 0  |
|        | <i>Anthophorula</i> | <i>Anthophorisca</i>    | <i>Anthophorula nitens</i>      | generalist[8]  | below ground[8] | solitary[8] | 0  | 0 | 0   | 1  |
|        |                     | <i>Anthophorula</i>     | <i>Anthophorula albicans</i>    | generalist[3]  | below ground[2] | solitary[2] | 0  | 0 | 1   | 0  |
|        |                     | <i>Anthophorula</i>     | <i>Anthophorula cerei</i>       | generalist[3]  |                 |             | 0  | 0 | 1   | 0  |
|        |                     | <i>Anthophorula</i>     | <i>Anthophorula torticornis</i> | generalist[3]  |                 |             | 0  | 0 | 1   | 1  |
|        | <i>Apis</i>         | <i>Apis</i>             | <i>Apis mellifera</i>           | generalist[2]  | cavity[2]       | eusocial[6] | 10 | 0 | 633 | 0  |
|        | <i>Bombus</i>       | <i>Bombias</i>          | <i>Bombus nevadensis</i>        | generalist[3]  | below ground[3] | eusocial[3] | 0  | 1 | 0   | 0  |
|        |                     | <i>Cullumanobombus</i>  | <i>Bombus crotchii</i>          |                |                 |             | 0  | 0 | 3   | 0  |
|        |                     | <i>Pyrobombus</i>       | <i>Bombus melanopygus</i>       |                |                 |             | 2  | 0 | 1   | 0  |
|        |                     | <i>Pyrobombus</i>       | <i>Bombus vandykei</i>          |                |                 |             | 0  | 0 | 1   | 0  |
|        |                     | <i>Pyrobombus</i>       | <i>Bombus vosnesenskii</i>      |                |                 |             | 1  | 0 | 10  | 1  |
|        |                     | <i>Thoracobombus</i>    | <i>Bombus californicus</i>      |                |                 |             | 1  | 3 | 4   | 4  |
|        |                     | <i>Thoracobombus</i>    | <i>Bombus sonorus</i>           |                |                 |             | 2  | 0 | 12  | 1  |
|        | <i>Ceratina</i>     | <i>Ceratinula</i>       | <i>Ceratina arizonensis</i>     | generalist[3]  | cavity[2]       | solitary[2] | 0  | 0 | 2   | 1  |
|        |                     | <i>Zadontomerus</i>     | <i>Ceratina acantha</i>         | generalist[3]  |                 |             | 9  | 3 | 40  | 0  |
|        |                     | <i>Zadontomerus</i>     | <i>Ceratina micheneri</i>       | generalist[3]  |                 |             | 0  | 0 | 1   | 0  |
|        |                     | <i>Zadontomerus</i>     | <i>Ceratina nanula</i>          | generalist[3]  |                 |             | 1  | 0 | 9   | 0  |
|        |                     | <i>Zadontomerus</i>     | <i>Ceratina sequoiae</i>        | specialist[3]  |                 |             | 0  | 0 | 1   | 0  |

|  |                   |                       |                               |                         |                         |                         |   |    |     |     |
|--|-------------------|-----------------------|-------------------------------|-------------------------|-------------------------|-------------------------|---|----|-----|-----|
|  |                   |                       |                               |                         |                         |                         |   |    |     |     |
|  | <i>Diadasia</i>   | <i>Coquillettapis</i> | <i>Diadasia angusticeps</i>   | specialist[9]           | below ground[2]         | solitary[9]             | 0 | 0  | 1   | 0   |
|  |                   | <i>Coquillettapis</i> | <i>Diadasia australis</i>     |                         |                         |                         | 1 | 1  | 20  | 26  |
|  |                   | <i>Coquillettapis</i> | <i>Diadasia bituberculata</i> |                         |                         |                         | 1 | 0  | 9   | 6   |
|  |                   | <i>Coquillettapis</i> | <i>Diadasia diminuta</i>      |                         |                         |                         | 0 | 5  | 3   | 3   |
|  |                   | <i>Coquillettapis</i> | <i>Diadasia laticauda</i>     |                         |                         |                         | 0 | 0  | 13  | 4   |
|  |                   | <i>Coquillettapis</i> | <i>Diadasia lutzi</i>         |                         |                         |                         | 0 | 0  | 1   | 0   |
|  |                   | <i>Coquillettapis</i> | <i>Diadasia nitidifrons</i>   |                         |                         |                         | 0 | 4  | 8   | 2   |
|  |                   | <i>Coquillettapis</i> | <i>Diadasia rinconis</i>      |                         |                         |                         | 0 | 0  | 125 | 42  |
|  |                   | <i>Coquillettapis</i> | <i>Diadasia vallicola</i>     |                         |                         |                         | 2 | 0  | 0   | 0   |
|  |                   | <i>Dasiapis</i>       | <i>Diadasia ochracea</i>      |                         |                         |                         | 1 | 0  | 22  | 0   |
|  |                   | <i>Diadasia</i>       | <i>Diadasia enavata</i>       |                         |                         |                         | 0 | 0  | 2   | 0   |
|  | <i>Eucera</i>     | <i>Synhalonia</i>     | <i>Eucera cordleyi</i>        | generalist[3]           | below ground[2]         | solitary[2]             | 0 | 0  | 1   | 3   |
|  |                   | <i>Synhalonia</i>     | <i>Eucera dorsata</i>         | generalist[3]           |                         |                         | 0 | 0  | 1   | 0   |
|  |                   | <i>Synhalonia</i>     | <i>Eucera lunata</i>          | generalist[3]           |                         |                         | 0 | 0  | 1   | 0   |
|  | <i>Habropoda</i>  | [NA]                  | <i>Habropoda depressa</i>     | generalist[10]          | below ground[10]        | solitary[10]            | 2 | 0  | 0   | 0   |
|  |                   |                       | <i>Habropoda tristissima</i>  | generalist[10]          | below ground[10]        | solitary[10]            | 0 | 0  | 8   | 0   |
|  | <i>Melecta</i>    | <i>Melecta</i>        | <i>Melecta pacifica</i>       | cleptoparasitic[2]<br>] | cleptoparasitic[2]<br>] | cleptoparasitic[2]<br>] | 0 | 0  | 1   | 0   |
|  | <i>Melissodes</i> | [unk]                 | <i>Melissodes</i> sp. 4       | generalist[2]           | below ground[2]         | solitary[11]            | 0 | 0  | 1   | 0   |
|  |                   | [unk]                 | <i>Melissodes</i> sp. A       |                         |                         |                         | 3 | 12 | 202 | 133 |
|  |                   | [unk]                 | <i>Melissodes</i> sp. B       |                         |                         |                         | 1 | 3  | 103 | 17  |
|  |                   | [unk]                 | <i>Melissodes</i> sp. C       |                         |                         |                         | 1 | 4  | 78  | 107 |
|  |                   | [unk]                 | <i>Melissodes</i> sp. D       |                         |                         |                         | 4 | 4  | 31  | 62  |
|  | <i>Nomada</i>     | [NA]                  | <i>Nomada</i> sp. A           | cleptoparasitic[2]      | cleptoparasitic[2]      | cleptoparasitic[2]      | 1 | 1  | 0   | 0   |

|            |                       |                      |                                  |                         |                         |                         |    |    |      |     |
|------------|-----------------------|----------------------|----------------------------------|-------------------------|-------------------------|-------------------------|----|----|------|-----|
|            |                       |                      |                                  | ]                       | ]                       | ]                       |    |    |      |     |
|            | <i>Oreopasites</i>    | [unk]                | <i>Oreopasites</i> sp. 1         | cleptoparasitic[2]<br>] | cleptoparasitic[2]<br>] | cleptoparasitic[2]<br>] | 0  | 0  | 0    | 1   |
|            |                       | [unk]                | <i>Oreopasites</i> sp. A         |                         |                         |                         | 0  | 0  | 3    | 0   |
|            | <i>Peponapis</i> *    | <i>Peponapis</i>     | <i>Peponapis pruinosa</i>        | specialist[2]           | below ground[7]         | solitary[12]            | 0  | 0  | 53   | 116 |
|            | <i>Svastra</i>        | <i>Epimelissodes</i> | <i>Svastra sabinensis</i>        | generalist[13]          | below<br>ground[14]     | solitary[13]            | 0  | 2  | 0    | 0   |
|            | <i>Triepeolus</i>     | [NA]                 | <i>Triepeolus concavus</i>       | cleptoparasitic[2]<br>] | cleptoparasitic[2]<br>] | cleptoparasitic[2]<br>] | 0  | 1  | 0    | 0   |
|            |                       |                      | <i>Triepeolus</i> sp. B          |                         |                         |                         | 2  | 1  | 0    | 0   |
|            | <i>Xenoglossa</i>     | <i>Eoxenoglossa</i>  | <i>Xenoglossa strenua</i>        | specialist[2]           | below ground[2]         | solitary[2]             | 0  | 0  | 72   | 103 |
|            |                       | <i>Xenoglossa</i>    | <i>Xenoglossa patricia</i>       |                         |                         |                         | 0  | 0  | 1    | 3   |
|            | <i>Xeromelecta</i>    | <i>Melectomorpha</i> | <i>Xeromelecta californica</i>   | cleptoparasitic[2]<br>] | cleptoparasitic[2]<br>] | cleptoparasitic[2]<br>] | 0  | 0  | 0    | 1   |
| Colletidae | <i>Colletes</i>       | [NA]                 | <i>Colletes</i> sp. A1           | generalist[3]           | below ground[2]         | solitary[2]             | 1  | 0  | 0    | 0   |
|            |                       |                      | <i>Colletes</i> sp. A2           |                         |                         |                         | 1  | 0  | 0    | 0   |
|            |                       |                      | <i>Colletes</i> sp. B            |                         |                         |                         | 3  | 0  | 0    | 0   |
|            | <i>Hylaeus</i>        | <i>Hylaeus</i>       | <i>Hylaeus mesillae</i>          | generalist[2]           | cavity[2]               | solitary[2]             | 2  | 12 | 2    | 1   |
| Halictidae | <i>Agapostemon</i>    | <i>Agapostemon</i>   | <i>Agapostemon femoratus</i>     | generalist[16]          | below ground[2]         | solitary[17]            | 0  | 0  | 28   | 1   |
|            |                       | <i>Agapostemon</i>   | <i>Agapostemon melliventris</i>  | generalist[16]          |                         | solitary[17]            | 1  | 1  | 274  | 16  |
|            |                       | <i>Agapostemon</i>   | <i>Agapostemon texanus</i>       | generalist[16]          |                         | solitary[18]            | 12 | 17 | 1692 | 112 |
|            | <i>Augochlorella</i>  | <i>Augochlorella</i> | <i>Augochlorella pomoniella</i>  | generalist[19]          | below<br>ground[20]     | eusocial[21]            | 1  | 2  | 22   | 3   |
|            | <i>Conanthalictus</i> | <i>Phaceliapis</i>   | <i>Conanthalictus cockerelli</i> | specialist[2]           | below<br>ground[22]     | solitary[23]            | 0  | 0  | 0    | 1   |
|            | <i>Dieunomia</i>      | <i>Epinomia</i>      | <i>Dieunomia nevadensis</i>      | generalist[24]          | below<br>ground[24]     | solitary[24]            | 1  | 1  | 0    | 0   |
|            | <i>Dufourea</i>       | [NA]                 | <i>Dufourea</i> sp. 2            | specialist[25]          | below                   | solitary[23]            | 1  | 0  | 8    | 2   |

|  |                     |                     |                               |                |                  |              |    |    |     |    |
|--|---------------------|---------------------|-------------------------------|----------------|------------------|--------------|----|----|-----|----|
|  |                     |                     | <i>Dufourea trochantera</i>   |                | ground[23]       |              | 0  | 1  | 308 | 39 |
|  | <i>Halictus</i>     | <i>Nealictus</i>    | <i>Halictus farinosus</i>     | generalist[26] | below ground[26] | eusocial[27] | 0  | 0  | 36  | 3  |
|  |                     | <i>Odontalictus</i> | <i>Halictus ligatus</i>       | generalist[28] | below ground[28] | eusocial[28] | 40 | 15 | 40  | 2  |
|  |                     | <i>Seladonia</i>    | <i>Halictus tripartitus</i>   | generalist[29] | below ground[30] | eusocial[30] | 9  | 0  | 198 | 0  |
|  | <i>Lasioglossum</i> | <i>Dialictus</i>    | <i>Lasioglossum</i> sp. dia1  | generalist[2]  | below ground[2]  | eusocial[2]  | 1  | 0  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia10 |                |                  |              | 1  | 0  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia11 |                |                  |              | 1  | 0  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia12 |                |                  |              | 1  | 0  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia2  |                |                  |              | 11 | 0  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia3  |                |                  |              | 0  | 4  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia4  |                |                  |              | 4  | 0  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia5  |                |                  |              | 1  | 0  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia6  |                |                  |              | 1  | 0  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia7  |                |                  |              | 2  | 1  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia8  |                |                  |              | 5  | 1  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. dia9  |                |                  |              | 1  | 0  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. diaA  |                |                  |              | 0  | 8  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. diaD  |                |                  |              | 0  | 1  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. diaE  |                |                  |              | 0  | 1  | 0   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. disA  |                |                  |              | 0  | 0  | 980 | 10 |
|  |                     |                     | <i>Lasioglossum</i> sp. disB  |                |                  |              | 0  | 0  | 66  | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. disD  |                |                  |              | 0  | 0  | 5   | 0  |
|  |                     |                     | <i>Lasioglossum</i> sp. disE  |                |                  |              | 0  | 0  | 333 | 0  |
|  |                     | <i>Evyllaenus</i>   | <i>Lasioglossum</i> sp. EV01  | generalist[2]  | below ground[2]  | eusocial[2]  | 0  | 0  | 208 | 3  |

|              |                       |                      |                                  |                    |                    |                    |   |   |    |   |
|--------------|-----------------------|----------------------|----------------------------------|--------------------|--------------------|--------------------|---|---|----|---|
|              |                       |                      | <i>Lasioglossum</i> sp. EV02     |                    |                    |                    | 5 | 0 | 19 | 0 |
|              |                       |                      | <i>Lasioglossum</i> sp. EV03     |                    |                    |                    | 1 | 0 | 42 | 1 |
|              |                       |                      | <i>Lasioglossum</i> sp. EV04     |                    |                    |                    | 0 | 0 | 3  | 0 |
|              |                       |                      | <i>Lasioglossum</i> sp. EV05     |                    |                    |                    | 0 | 0 | 1  | 0 |
|              |                       |                      | <i>Lasioglossum</i> sp. EV06     |                    |                    |                    | 1 | 0 | 3  | 0 |
|              |                       |                      | <i>Lasioglossum</i> sp. EV07     |                    |                    |                    | 0 | 0 | 2  | 0 |
|              | <i>Lasioglossum</i>   |                      | <i>Lasioglossum</i> sp. LA02     | generalist[2]      | below ground[2]    | eusocial[2]        | 0 | 0 | 3  | 0 |
|              |                       |                      | <i>Lasioglossum sisymbrii</i>    | generalist[31]     | below ground[32]   | solitary[33]       | 4 | 0 | 49 | 0 |
|              | <i>Micralictoides</i> | [NA]                 | <i>Micralictoides ruficaudus</i> | specialist[34]     | below ground[35]   | solitary[35]       | 4 | 1 | 4  | 1 |
|              | <i>Sphecodes</i>      | [NA]                 | <i>Sphecodes</i> sp. A           | cleptoparasitic[2] | cleptoparasitic[2] | cleptoparasitic[2] | 1 | 1 | 0  | 1 |
| Megachilidae | <i>Anthidium</i>      | <i>Anthidium</i>     | <i>Anthidium placitum</i>        | specialist[36]     | below ground[36]   | solitary[36]       | 0 | 0 | 1  | 0 |
|              | <i>Ashmeadiella</i>   | <i>Ashmeadiella</i>  | <i>Ashmeadiella buconis</i>      | generalist[37]     | cavity[37]         | solitary[6]        | 9 | 2 | 0  | 0 |
|              |                       | <i>Ashmeadiella</i>  | <i>Ashmeadiella cubiceps</i>     | generalist[3]      |                    |                    | 1 | 0 | 0  | 0 |
|              |                       | <i>Ashmeadiella</i>  | <i>Ashmeadiella difugita</i>     | generalist[3]      |                    |                    | 2 | 0 | 0  | 0 |
|              |                       | <i>Ashmeadiella</i>  | <i>Ashmeadiella meliloti</i>     | generalist[3]      |                    |                    | 1 | 0 | 0  | 0 |
|              |                       | <i>Ashmeadiella</i>  | <i>Ashmeadiella</i> sp. 1        | generalist[3]      |                    |                    | 0 | 0 | 0  | 1 |
|              |                       | <i>Cubitognatha</i>  | <i>Ashmeadiella salviae</i>      | specialist[37]     |                    |                    | 2 | 0 | 0  | 0 |
|              | <i>Atoposmia</i>      | <i>Eremosmia</i>     | <i>Atoposmia</i> sp. A           | specialist[3]      | cavity[3]          | solitary[6]        | 0 | 0 | 1  | 0 |
|              | <i>Dianthidium</i>    | <i>Dianthidium</i>   | <i>Dianthidium parvum</i>        | generalist[3]      | cavity[3]          | solitary[6]        | 0 | 0 | 1  | 0 |
|              |                       | <i>Dianthidium</i>   | <i>Dianthidium pudicum</i>       | generalist[3]      | cavity[3]          |                    | 1 | 0 | 1  | 0 |
|              | <i>Hoplitis</i>       | [unk]                | <i>Hoplitis</i> sp. 1            | generalist[3]      | cavity[2]          | solitary[6]        | 0 | 0 | 1  | 0 |
|              |                       | [unk]                | <i>Hoplitis</i> sp. 2            |                    |                    |                    | 0 | 0 | 4  | 2 |
|              | <i>Megachile</i>      | <i>Litomegachile</i> | <i>Megachile brevis</i>          | generalist[38]     | cavity[38]         | solitary[38]       | 1 | 0 | 0  | 0 |
|              |                       | <i>Sayapis</i>       | <i>Megachile inimca</i>          | generalist[3]      | cavity[2]          | solitary[2]        | 2 | 0 | 0  | 0 |

|  |              |       |                         |               |           |             |   |   |   |   |
|--|--------------|-------|-------------------------|---------------|-----------|-------------|---|---|---|---|
|  |              |       |                         | generalist[3] | cavity[2] | solitary[2] |   |   |   |   |
|  |              | [unk] | <i>Megachile</i> sp. 2  |               |           |             | 0 | 1 | 0 | 0 |
|  |              | [unk] | <i>Megachile</i> sp. 3a |               |           |             | 0 | 1 | 0 | 0 |
|  |              | [unk] | <i>Megachile</i> sp. 3b |               |           |             | 0 | 1 | 0 | 0 |
|  |              | [unk] | <i>Megachile</i> sp. 4  |               |           |             | 0 | 0 | 0 | 1 |
|  |              | [unk] | <i>Megachile</i> sp. 5  |               |           |             | 0 | 0 | 0 | 1 |
|  |              | [unk] | <i>Megachile</i> sp. D  |               |           |             | 1 | 0 | 0 | 0 |
|  |              | [unk] | <i>Megachile</i> sp. E  |               |           |             | 1 | 0 | 2 | 0 |
|  |              | [unk] | <i>Megachile</i> sp. F  |               |           |             | 1 | 0 | 0 | 0 |
|  |              | [unk] | <i>Megachile</i> sp. G  |               |           |             | 0 | 0 | 2 | 0 |
|  |              | [unk] | <i>Megachile</i> sp. H  |               |           |             | 0 | 0 | 3 | 0 |
|  |              | [unk] | <i>Megachile</i> sp. I  |               |           |             | 0 | 0 | 1 | 0 |
|  |              | [unk] | <i>Megachile</i> sp. J  |               |           |             | 0 | 0 | 1 | 0 |
|  |              | [unk] | <i>Megachile</i> sp. X  |               |           |             | 1 | 4 | 0 | 0 |
|  | <i>Osmia</i> | [unk] | <i>Osmia</i> sp. 1      | generalist[3] | cavity[2] | solitary[2] | 0 | 1 | 0 | 0 |
|  |              | [unk] | <i>Osmia</i> sp. A      |               |           |             | 3 | 0 | 0 | 0 |
|  |              | [unk] | <i>Osmia</i> sp. C      |               |           |             | 1 | 0 | 0 | 0 |
|  |              | [unk] | <i>Osmia</i> sp. D      |               |           |             | 1 | 0 | 0 | 0 |
|  |              | [unk] | <i>Osmia</i> sp. E      |               |           |             | 1 | 0 | 0 | 0 |
|  |              | [unk] | <i>Osmia</i> sp. F      |               |           |             | 2 | 0 | 0 | 0 |
|  |              | [unk] | <i>Osmia</i> sp. G      |               |           |             | 0 | 0 | 2 | 0 |
|  |              | [unk] | <i>Osmia</i> sp. H      |               |           |             | 0 | 0 | 1 | 0 |

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**Table S4.** Results of significance tests for fixed effects, including interaction terms, from mixed models with nursery and year as random effects. Symbols in cells indicate the results of that hypothesis test: \*\*\*:  $P \leq 0.001$ ; \*\*:  $P \leq 0.01$ ; \*:  $P \leq 0.05$ ; ns:  $P > 0.05$ . Parentheses around “ns” indicate that, in the identical model with *Apis mellifera* included, the  $P$ -value for that hypothesis test was  $\leq 0.05$ . Functional dispersion (FDis) accounts for multiple functional traits, so the two models for this variable lacked a functional trait effect and associated interactions (“N/A”).

| bee collection method | independent variable          | numerator degrees of freedom | proportions of <b>individuals</b> |                  |                     | proportions of <b>species</b> |                  |                     | functional dispersion (FDis) |
|-----------------------|-------------------------------|------------------------------|-----------------------------------|------------------|---------------------|-------------------------------|------------------|---------------------|------------------------------|
|                       |                               |                              | diet breadth                      | nesting location | social organization | diet breadth                  | nesting location | social organization |                              |
| sweep netting         | functional trait              | 2                            | ***                               | **               | ***                 | ***                           | **               | *                   | N/A                          |
|                       | season                        | 2                            | ns                                | *                | *                   | ns                            | ns               | ns                  | ns                           |
|                       | nursery type                  | 1                            | **                                | ns               | ns                  | *                             | ns               | ns                  | ns                           |
|                       | prop. natural area at 1 km    | 1                            | ns                                | ns               | ns                  | ns                            | ns               | ns                  | ns                           |
|                       | nursery area                  | 1                            | ns                                | ns               | ns                  | ns                            | ns               | ns                  | ns                           |
|                       | trait x season                | 4                            | ns                                | **               | *                   | ns                            | ns               | ns                  | N/A                          |
|                       | trait x nursery type          | 2                            | ***                               | ns               | ns                  | **                            | ns               | ns                  | N/A                          |
|                       | season x nursery type         | 2                            | ns                                | ns               | ns                  | ns                            | ns               | ns                  | ns                           |
|                       | trait x season x nursery type | 4                            | ns                                | ns               | ns                  | ns                            | ns               | ns                  | N/A                          |
| vane trapping         | functional trait              | 2                            | ***                               | ***              | ***                 | ***                           | ***              | ***                 | N/A                          |
|                       | season                        | 2                            | ***                               | (ns)             | ***                 | ns                            | ns               | ns                  | ns                           |
|                       | nursery type                  | 1                            | ns                                | ***              | ***                 | ns                            | ns               | ns                  | ns                           |
|                       | prop. natural area at 1 km    | 1                            | ns                                | ns               | ns                  | ns                            | ns               | ns                  | ns                           |
|                       | nursery area                  | 1                            | ns                                | ns               | ns                  | ns                            | ns               | ns                  | ns                           |
|                       | trait x season                | 4                            | ***                               | (ns)             | ***                 | *                             | ns               | ns                  | N/A                          |
|                       | trait x nursery type          | 2                            | (ns)                              | ***              | ***                 | *                             | *                | ns                  | N/A                          |
|                       | season x nursery type         | 2                            | ***                               | *                | ***                 | ns                            | ns               | ns                  | *                            |
|                       | trait x season x nursery type | 4                            | ***                               | **               | ***                 | ns                            | ns               | ns                  | N/A                          |



**Table S6.** Numbers of bee species detected via sweep netting and vane trapping, categorized by three functional traits of interest: diet breadth, nesting location, and social organization. Cleptoparasites are tallied separately.

| <b>Sweep netting (92 spp.)</b>  |              | eusocial | solitary |
|---------------------------------|--------------|----------|----------|
| generalist                      | below ground | 26       | 22       |
|                                 | cavity       | 0        | 25       |
| specialist                      | below ground | 0        | 14       |
|                                 | cavity       | 0        | 1        |
| cleptoparasites: 4              |              |          |          |
| <b>Vane trapping (102 spp.)</b> |              | eusocial | solitary |
| generalist                      | below ground | 22       | 22       |
|                                 | cavity       | 0        | 20       |
| specialist                      | below ground | 0        | 31       |
|                                 | cavity       | 0        | 2        |
| cleptoparasites: 5              |              |          |          |

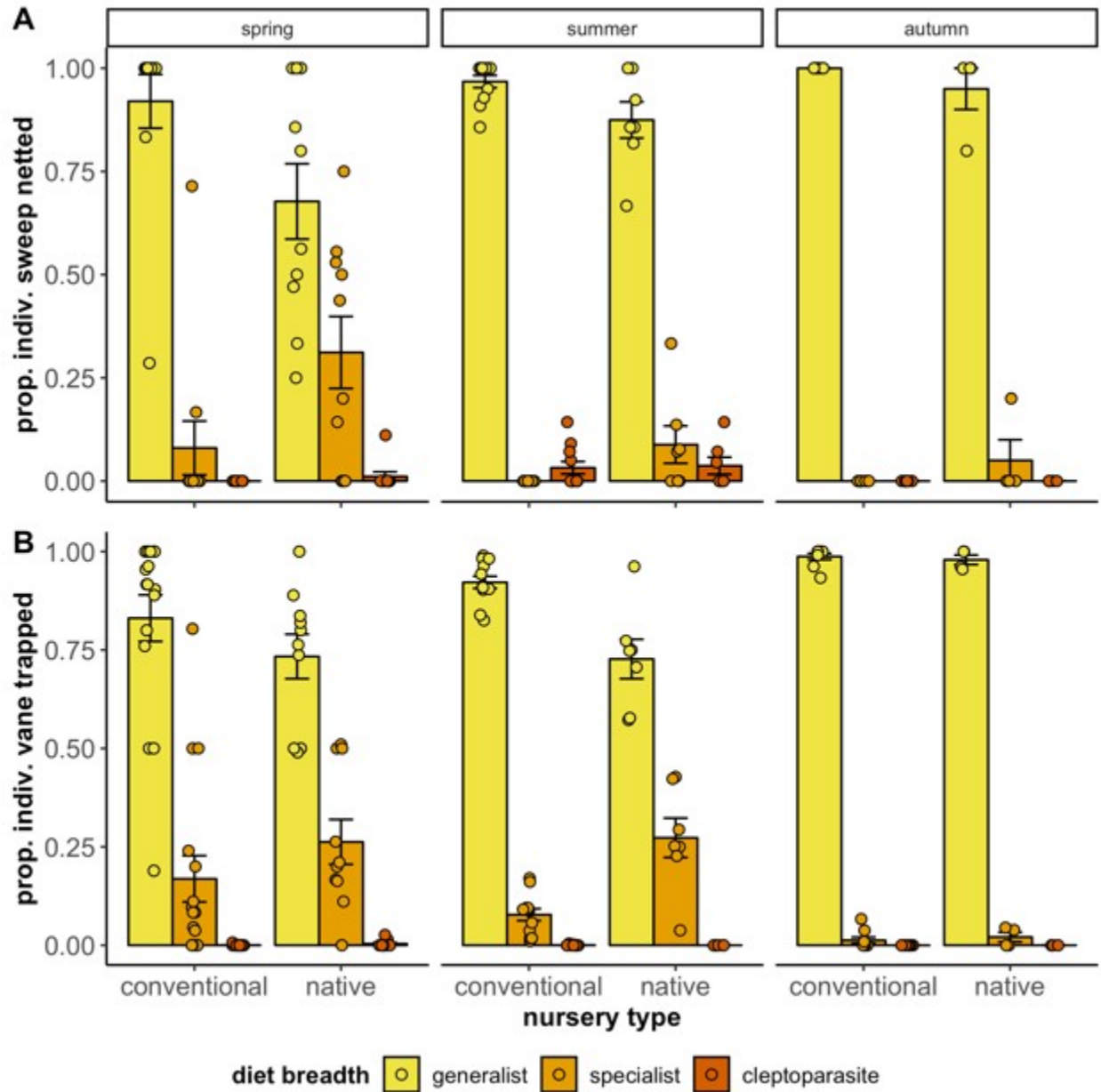
**Table S7.** List of flowering plant species off which bees were collected, organized by their native status ( $N = 375$  bees). Shown are the numbers of bees collected off each plant species at native and at conventional plant nurseries, respectively (note that a “0” does *not* necessarily imply that plant species was present at that type of nursery).

| <b>plant species name</b>         | <b>California native status</b> | <b>native nurseries</b> | <b>conventional nurseries</b> |
|-----------------------------------|---------------------------------|-------------------------|-------------------------------|
| <i>Salvia clevelandii</i>         | native                          | 5                       | 25                            |
| <i>Encelia californica</i>        | native                          | 12                      | 8                             |
| <i>Baileya multiradiata</i>       | native                          | 14                      | 0                             |
| <i>Erigeron glaucus</i>           | native                          | 10                      | 3                             |
| <i>Eschscholzia californica</i>   | native                          | 13                      | 0                             |
| <i>Abutilon palmeri</i>           | native                          | 10                      | 1                             |
| <i>Eriophyllum confertiflorum</i> | native                          | 2                       | 7                             |
| <i>Salvia mellifera</i>           | native                          | 8                       | 1                             |
| <i>Ceanothus</i> sp.              | native                          | 8                       | 0                             |
| <i>Grindelia camporum</i>         | native                          | 8                       | 0                             |
| <i>Penstemon heterophyllus</i>    | native                          | 1                       | 6                             |
| <i>Baccharis pilularis</i>        | native                          | 0                       | 6                             |
| <i>Corethrogyne filaginifolia</i> | native                          | 5                       | 1                             |
| <i>Dendromecon harfordii</i>      | native                          | 6                       | 0                             |
| <i>Eriogonum fasciculatum</i>     | native                          | 3                       | 3                             |
| <i>Penstemon spectabilis</i>      | native                          | 4                       | 2                             |
| <i>Trichostema lanatum</i>        | native                          | 6                       | 0                             |
| <i>Malosma laurina</i>            | native                          | 5                       | 0                             |
| <i>Sphaeralcea ambigua</i>        | native                          | 5                       | 0                             |
| <i>Baccharis salicifolia</i>      | native                          | 3                       | 1                             |
| <i>Rosa californica</i>           | native                          | 3                       | 1                             |
| <i>Encelia farinosa</i>           | native                          | 0                       | 3                             |
| <i>Epilobium canum</i>            | native                          | 3                       | 0                             |
| <i>Eriogonum parvifolium</i>      | native                          | 3                       | 0                             |
| <i>Fallugia paradoxa</i>          | native                          | 3                       | 0                             |
| <i>Helianthus annuus</i>          | native                          | 3                       | 0                             |
| <i>Isocoma menziesii</i>          | native                          | 1                       | 2                             |
| <i>Achillea millefolium</i>       | native                          | 0                       | 2                             |
| <i>Astragalus trichopodus</i>     | native                          | 2                       | 0                             |
| <i>Chilopsis linearis</i>         | native                          | 0                       | 2                             |
| <i>Clarkia unguiculata</i>        | native                          | 2                       | 0                             |
| <i>Frangula californica</i>       | native                          | 0                       | 2                             |
| <i>Grindelia stricta</i>          | native                          | 2                       | 0                             |
| <i>Phyla nodiflora</i>            | native                          | 0                       | 2                             |
| <i>Sidalcea malviflora</i>        | native                          | 2                       | 0                             |
| <i>Tetrandeum acaulis</i>         | native                          | 2                       | 0                             |
| <i>Bahiopsis laciniata</i>        | native                          | 1                       | 0                             |
| <i>Calystegia macrostegia</i>     | native                          | 1                       | 0                             |
| <i>Dichelostemma capitatum</i>    | native                          | 1                       | 0                             |
| <i>Eriogonum grande</i>           | native                          | 1                       | 0                             |
| <i>Helianthus californicus</i>    | native                          | 1                       | 0                             |
| <i>Malacothamnus clementinus</i>  | native                          | 1                       | 0                             |
| <i>Malacothamnus fasciculatus</i> | native                          | 1                       | 0                             |
| <i>Monardella odoratissima</i>    | native                          | 1                       | 0                             |
| <i>Monardella villosa</i>         | native                          | 1                       | 0                             |

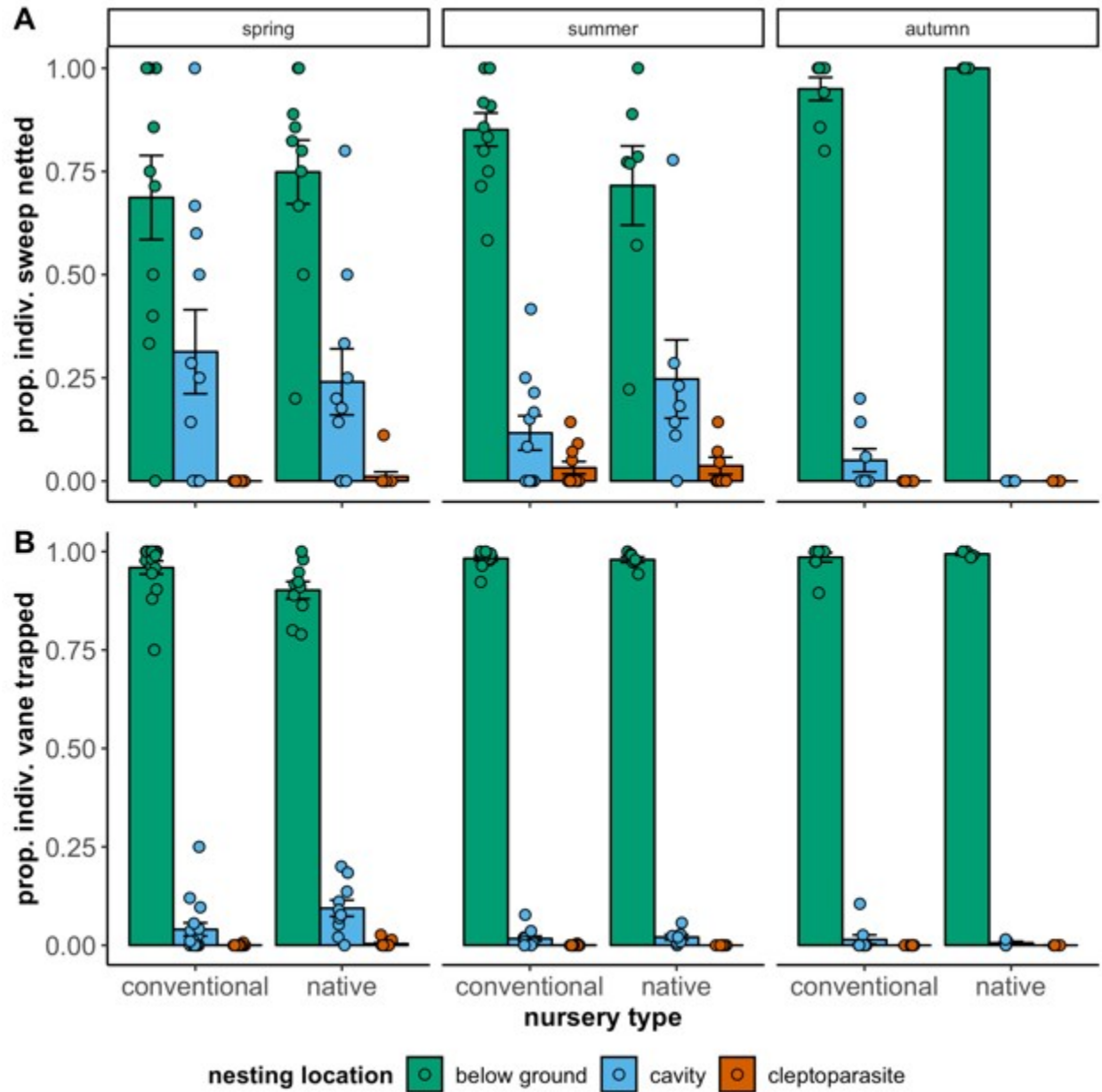
|                                    |            |   |   |
|------------------------------------|------------|---|---|
| <i>Romneya coulteri</i>            | native     | 0 | 1 |
| <i>Salvia leucophylla</i>          | native     | 0 | 1 |
| <i>Salvia pachyphylla</i>          | native     | 0 | 1 |
| <i>Sambucus mexicana</i>           | native     | 0 | 1 |
| <i>Salvia greggii</i>              | non-native | 0 | 9 |
| <i>Gaillardia grandiflora</i>      | non-native | 0 | 8 |
| <i>Lagerstroemia indica</i>        | non-native | 0 | 8 |
| <i>Eremophila hygrophana</i>       | non-native | 0 | 6 |
| <i>Salvia chamaedryoides</i>       | non-native | 0 | 6 |
| <i>Caryopteris clandonensis</i>    | non-native | 0 | 5 |
| <i>Melampodium leucanthum</i>      | non-native | 1 | 4 |
| <i>Vitex agnus-castus</i>          | non-native | 0 | 5 |
| <i>Bulbine frutescens</i>          | non-native | 0 | 4 |
| <i>Ligustrum lucidum</i>           | non-native | 0 | 4 |
| <i>Oenothera cf. biennis</i>       | non-native | 0 | 4 |
| <i>Rosmarinus officinalis</i>      | non-native | 0 | 4 |
| <i>Berlandiera lyrata</i>          | non-native | 3 | 0 |
| <i>Cistus salviifolius</i>         | non-native | 0 | 3 |
| <i>Lamium amplexicaule</i>         | non-native | 3 | 0 |
| <i>Oenothera speciosa</i>          | non-native | 0 | 3 |
| <i>Pelargonium sidoides</i>        | non-native | 0 | 3 |
| <i>Salvia microphylla</i>          | non-native | 1 | 2 |
| <i>Salvia uliginosa</i>            | non-native | 0 | 3 |
| <i>Tecoma capensis</i>             | non-native | 0 | 3 |
| <i>Verbesina encelioides</i>       | non-native | 0 | 3 |
| <i>Calandrinia spectabilis</i>     | non-native | 0 | 2 |
| <i>Coreopsis grandiflora</i>       | non-native | 0 | 2 |
| <i>Duranta erecta</i>              | non-native | 0 | 2 |
| <i>Lavandula dentata</i>           | non-native | 0 | 2 |
| <i>Lavandula stoechas</i>          | non-native | 0 | 2 |
| <i>Leptospermum scoparium</i>      | non-native | 0 | 2 |
| <i>Leucophyllum frutescens</i>     | non-native | 0 | 2 |
| <i>Perovskia atriplicifolia</i>    | non-native | 0 | 2 |
| <i>Ageratina adenophora</i>        | non-native | 0 | 1 |
| <i>Buddleja davidii</i>            | non-native | 0 | 1 |
| <i>Callistemon viminalis</i>       | non-native | 0 | 1 |
| <i>Ceratostigma plumbaginoides</i> | non-native | 0 | 1 |
| <i>Cercidium</i> sp.               | non-native | 0 | 1 |
| <i>Cistus purpureus</i>            | non-native | 0 | 1 |
| <i>Convolvulus sabatius</i>        | non-native | 0 | 1 |
| <i>Cosmos bipinnatus</i>           | non-native | 0 | 1 |
| <i>Felicia amelloides</i>          | non-native | 0 | 1 |
| <i>Gaura lindheimeri</i>           | non-native | 0 | 1 |
| <i>Gnaphalieae</i> sp.             | non-native | 0 | 1 |
| <i>Lavandula gingsii</i>           | non-native | 0 | 1 |
| <i>Monarda</i> sp.                 | non-native | 0 | 1 |
| <i>Nerium oleander</i>             | non-native | 0 | 1 |
| <i>Pallenis maritima</i>           | non-native | 0 | 1 |
| <i>Penstemon cf. cobaea</i>        | non-native | 0 | 1 |
| <i>Sagina subulata</i>             | non-native | 0 | 1 |
| <i>Salvia farinacea</i>            | non-native | 0 | 1 |
| <i>Vitex trifolia</i>              | non-native | 0 | 1 |



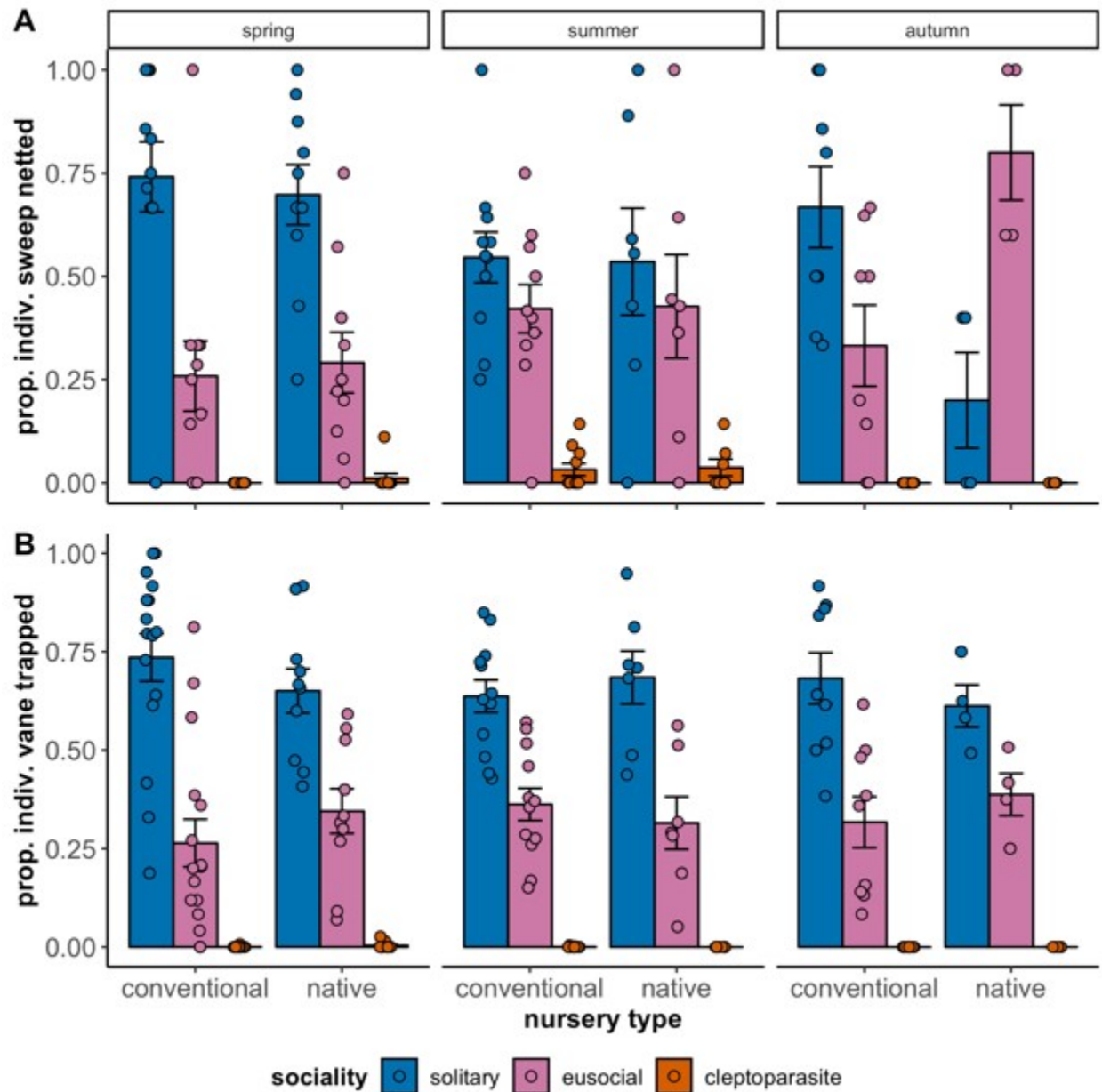
**Fig. S1.** View of a sampling plot with blue vane trap deployed at a conventional plant nursery in Riverside County, California, on 21 June 2016.



**Fig. S2.** Diet breadth of bee individuals collected by (A) sweep netting and (B) vane trapping. Proportions represent the fraction of individual bees detected a given sampling that fall into each of three functional categories—generalist, specialist, or cleptoparasite—out of all individuals collected in that given sampling. Bars and whiskers represent mean $\pm$ SE. Samplings are arranged by season and type of nursery. For sweep netting (A), the two-way interaction between diet breadth and nursery type was statistically significant. For vane trapping (B), the three-way interaction between diet breadth, season, and nursery type was statistically significant (Table S4).



**Fig. S3.** Nesting location of bee individuals collected by (A) sweep netting and (B) vane trapping. Proportions represent the fraction of individual bees detected a given sampling that fall into each of three functional categories— below ground, cavity, or cleptoparasite—out of all individuals collected in that given sampling. Bars and whiskers represent mean $\pm$ SE. Samplings are arranged by season and type of nursery. For sweep netting (A), the two-way interaction between nesting location and season was statistically significant. For vane trapping (B), the three-way interaction between diet breadth, season, and nursery type was statistically significant (Table S4).



**Fig. S4.** Social organization of bee individuals collected by (A) sweep netting and (B) vane trapping. Proportions represent the fraction of individual bees detected a given sampling that fall into each of three functional categories—solitary, eusocial, or cleptoparasite—out of all individuals collected in that given sampling. Bars and whiskers represent mean $\pm$ SE. Samplings are arranged by season and type of nursery. For sweep netting (A), the two-way interaction between sociality and season was statistically significant. For vane trapping (B), the three-way interaction between diet breadth, season, and nursery type was statistically significant (Table S4).